

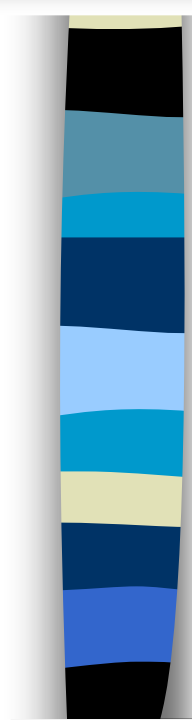
I keep bumping into people on my side

(Med & Sur) – Endocrinology

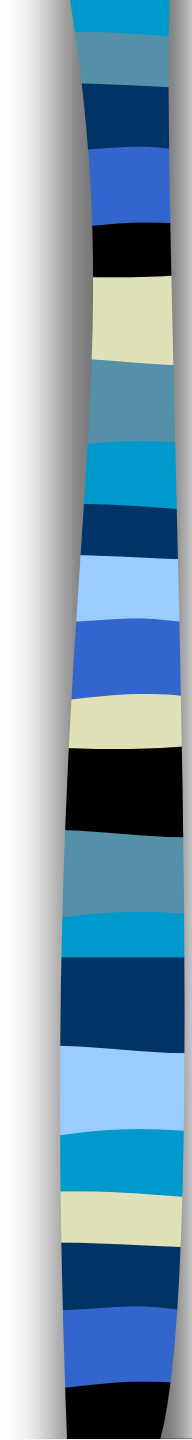
November, 2023



Prof. Karen S. L. Lam (Medicine)

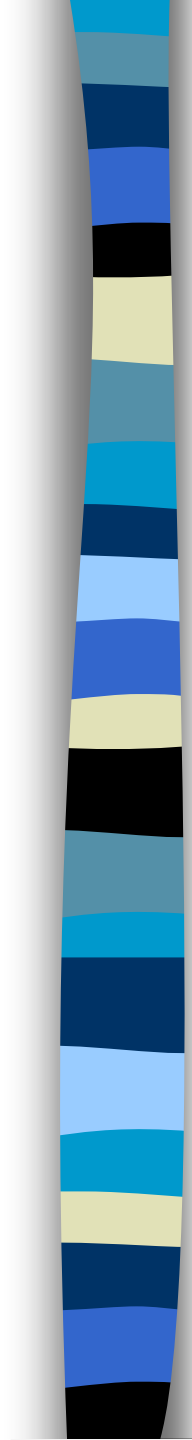


Prof. Gilberto Leung (Surgery)



Presentations of Pituitary Tumours

1. Local symptoms - visual loss, headache
2. S/S of pituitary hormone hyper-secretion and hypo-secretion
3. Incidental finding on skull X-ray, CT scan or MRI



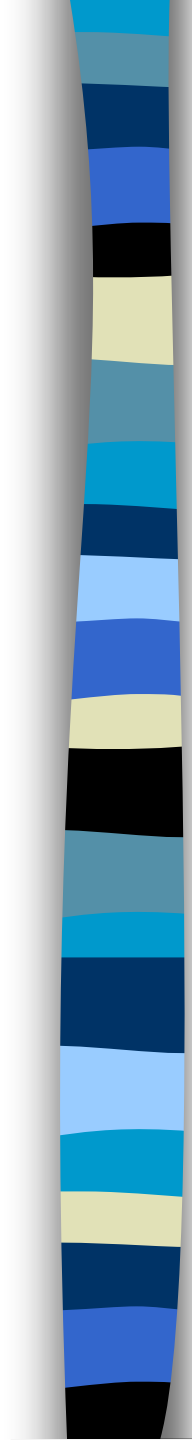
Types of Pituitary Tumours

Pituitary adenomas – more in adults

Also known as pituitary neuroendocrine tumours (PitNET): new name proposed by pathologists, but not widely supported by endocrinologists

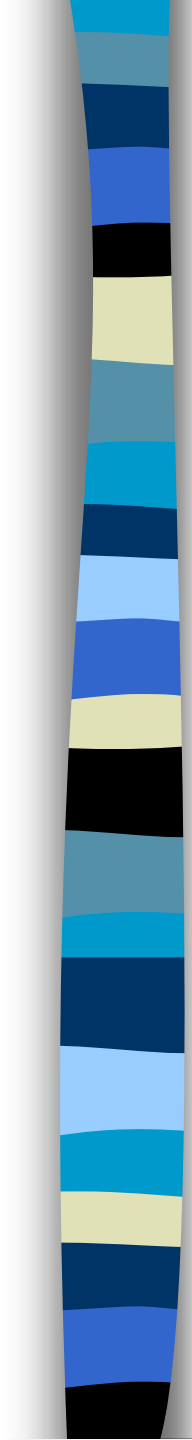
Craniopharyngiomas – more in children

Pituitary carcinomas - v rare; primary or secondary



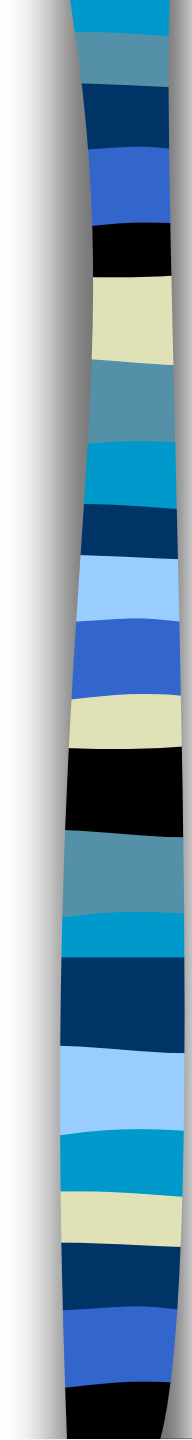
Craniopharyngiomas

- 50%: calcification present
- 50%: symptoms in childhood
- 25%: symptoms between 20-40 years
- Hypopituitarism e.g. growth retardation
- Stalk compression - diabetes insipidus
- Chiasmal compression - common
- Obstruction to 3rd ventricle may occur



Pituitary Adenomas

- 6-18% of all brain tumours
- Unselected autopsy series: prevalence 12-22%; most common being prolactinomas



Clinical Classification	Presentation
Functioning adenomas (70-80%)	
Prolactinomas	Galactorrhoea-amenorrhoea; impot
Growth hormone producing tumours	Acromegaly / gigant
ACTH producing tumours	Cushing's syndrome
Glycoprotein secreting tumours - TSH, FSH, LH, α - or β -subunits	2° hyperthyroidism - Hypopituitarism / pi mass / visual loss
Non-functioning adenomas (20-30%)	Hypopituitarism / pi mass / visual loss

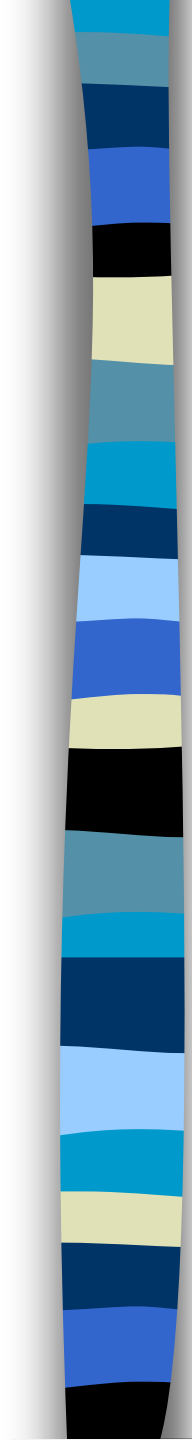
Pathological Classification According to Cell lineage (WHO 2022)

Transcription factors (or sequence-specific DNA-binding factor) are proteins that control the rate of transcription of genetic information from DNA to messenger RNA by binding to a specific DNA sequence. The major transcription factors which regulate pituitary cytogenesis, hormonal gene expression and hence, cell lineage include

- **TPIT** (T-box transcription factor; TBX19) - corticotroph differentiation
- **Pit1** (pituitary-specific transcription factor 1) - lactotroph, mammosomatotroph, somatotroph and thyrotroph differentiation
- **SF1** (steroidogenic factor 1) – gonadotroph differentiation

Advantages of the new classification:

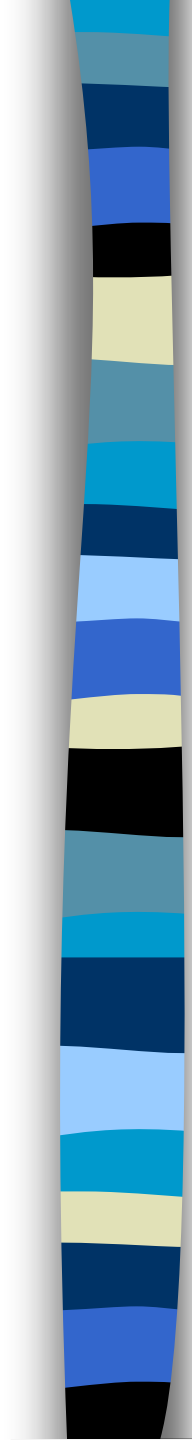
1. Improves accuracy of diagnosis of pituitary tumours: Immunohistochemical staining for transcription factors produces strongly positive nuclear staining (vs patchy cytoplasmic staining when using only hormonal staining). Followed by targeted hormone staining.
2. Prognostic implications: eg. SF1-lineage pituitary tumours run a relatively benign course; in those with complete resection post operatively, recurrence is rare



Other Causes of Hyperprolactinaemia

(non-stressed prolactin usually <3x normal)

Physiological	Pathological	Others
Pregnancy Lactation	Lesions (e.g. Tumours) involving the hypothalamus or pituitary stalk Oestrogen-containing medications Drugs affecting central dopaminergic regulation: Anti-psychotic drugs Anti-emetics α -methyldopa H2 antagonists Idiopathic	Hypothyroidism: increased TRH (thyrotropin releasing hormone) Renal failure: impaired dopamine action; reduced prolactin clearance Chest Injury



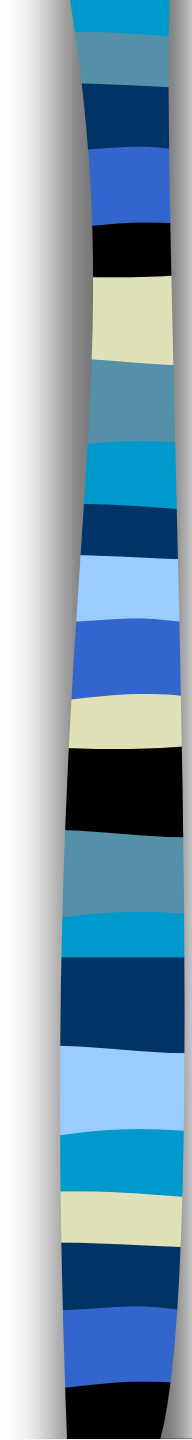
Diagnosis

Growth hormone (GH) -Producing Tumours

- (a) **Screening: serum insulin-like growth factor 1(IGF1):**
above normal for age and gender
- (b) **Confirmation: Non-suppressible serum GH**
during oral glucose tolerance test (OGTT):
 - Nadir > 0.4 ng/ml (1 mU/L)*[American Association of Clinical Endocrinologists 2011]*
 - Nadir > 1 ng/ml *[American Endocrine Society 2014]*

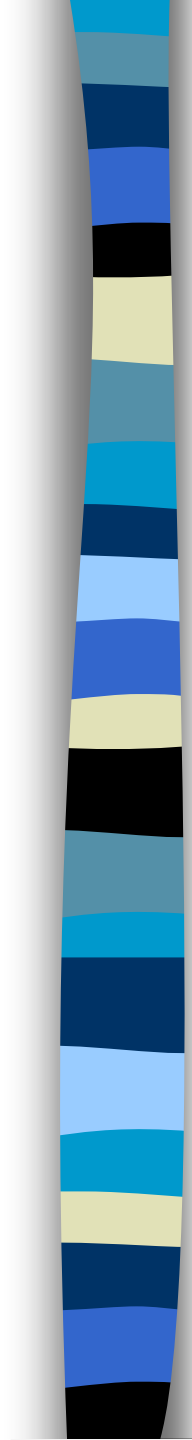
Prolactinomas

- (a) $\uparrow$ Prolactin; usually > 10 x normal
- (b) Tumour shrinkage by $>50\%$ in 90% patients when treated with dopamine agonists



Radiological Diagnosis of Pituitary Tumours

- MRI (magnetic resonance imaging):
imaging modality of choice



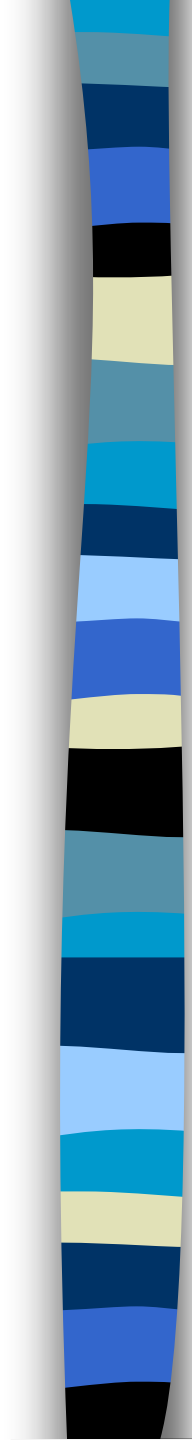
Treatment of Pituitary Tumours

1. Surgery

- All tumours causing pituitary hyper-function
- All macroadenomas
- For prolactinomas, medical treatment may be preferred
- Transsphenoidal: route of choice

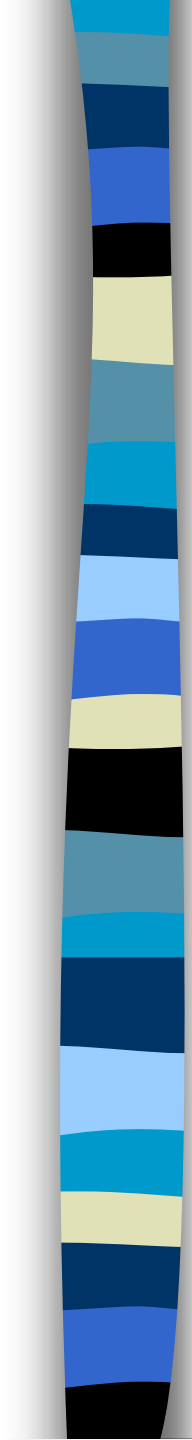
Approach: transnasal endoscopic or sub-labial

- Transfrontal: very large suprasellar extension, severe chiasmal compression
- Disadvantages : residual/recurrence esp. if macroadenomas; hypopituitarism, diabetes insipidus.
Acute complications: post-operative bleeding; CSF rhinorrhoea
- Advantage : rapid ↓ in secretion and tumour size



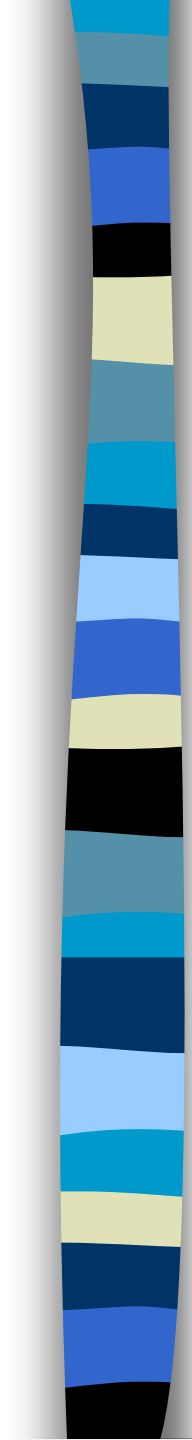
2. *Radiation therapy*

- Conventional external irradiation
- Radiosurgery: "Gamma knife"; "X-knife" (not used if suprasellar extension to within 5 mm of optic chiasm)
- Disadvantages - delayed effect on secretion
 - higher incidence of hypopituitarism
- Advantage - restrains tumour growth
- Useful adjunct to surgery
- May be primary therapy for macroprolactinomas



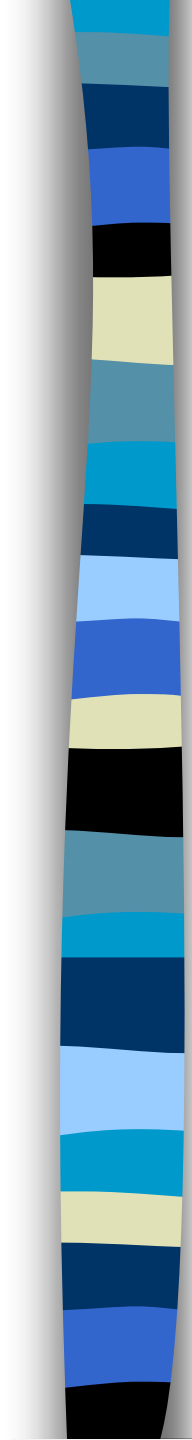
3. *Medical Treatment*

- Effects usually reversible on drug withdrawal
- Adjunct to surgery/RT except for prolactinomas for which this may be the sole treatment



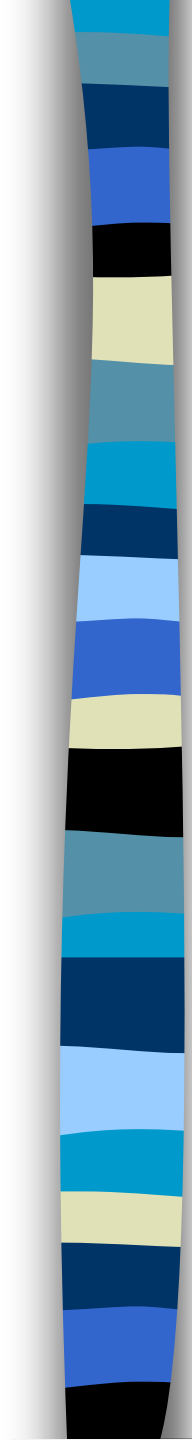
Bromocriptine & other dopamine agonists

- Prolactinomas: >90% patients achieve normal PRL and reduction in size on dopamine agonists
- Acromegaly: IGF-1 normalised in 10% patients in meta-analysis; no size reduction
- FSH producing tumours: reduce FSH but not tumour size
- Side-effects: Nausea, vomiting, orthostatic hypotension, headache, dizziness. Usually much less with cabergoline
- Cabergoline versus bromocriptine: less frequent dosing; more convenient; more effective; much higher cost. Screen for clinical valvular disease if high total dose such as $\geq 300\text{mg}$



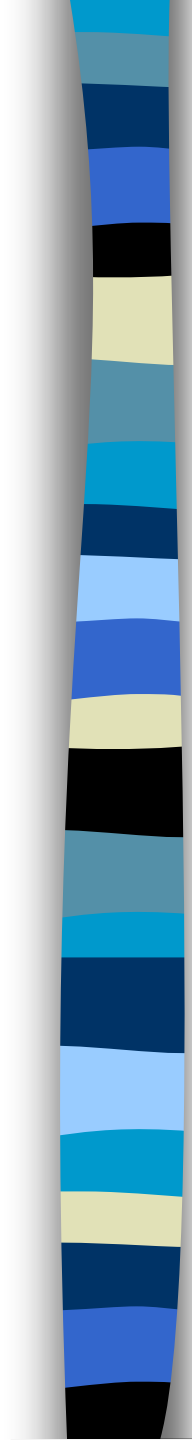
Somatostatin Analogs

- Synthetic somatostatin receptors (sstr) agonists
- Long acting analogs IMI every 4-6 weeks.
- Octreotide LAR; Lanreotide: Predominantly act on sstr2;
↓ secretion of GH & TSH producing tumours;
↓ size in 50% of patients
- For acromegaly, 60% of patients achieve normal GH/IGF1.
- Pasireotide LAR: high affinity for sstr5; effective for ACTH producing tumours & more effective for some GH producing tumours.
- Oral Octreotide capsules twice daily: FDA approved June 2020
- Oral non-peptide selective sstr2 agonist: clinical trial
- Useful adjunct to surgery & RT
- Side-effects: gall stone formation; may affect glucose tolerance (hyperglycaemia is common with Pasireotide)



Growth hormone receptor antagonist & acromegaly

- Pegvisomant – blocks GH receptor
- Daily injections – normal IGF1: 90% in 12 month study (Lancet 2001); 76% in long-term study: compliance problem (Eur J Endocrinol 2007)
- No ↓ in tumour size (monitor for ↑)
- Side-effect: ↑ liver transaminases (5% patients)
- Available only in some countries; high cost



Causes of Hypopituitarism

1. Structural damage of hypothalamus, pituitary or pituitary stalk

Tumours - large pituitary/hypothalamic tumours

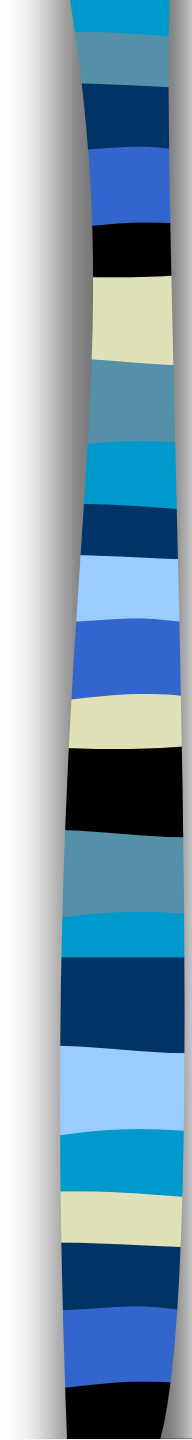
Trauma - surgery, radiation therapy, head injury

Infarction - post partum necrosis (Sheehan's Syndrome); pituitary apoplexy

Infiltration - haemosiderosis or haemochromatosis, histiocytosis, sarcoidosis

Infections - TB, syphilis, mycosis, toxoplasmosis (AIDS)

Immunologic - lymphocytic hypophysitis (may be side-effect of immune check-point [CTLA-4; PD-1] inhibitors)
- isolated ACTH deficiency



Causes of Hypopituitarism

II. Functional disorders: isolated/multiple

Genetic –

Panhypopituitarism

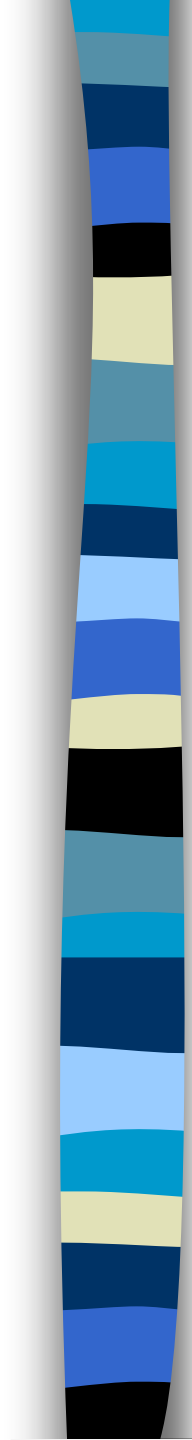
Isolated: GH deficiency

Isolated LH/FSH deficiency (Kallman's syndrome:
may have anosmia)

Reversible –

Emotional deprivation (GH)

Anorexia nervosa (LH/FSH $\pm$ TSH)



Practical Considerations

- Usu. no hypopituitarism if microadenoma (<1 cm)
- Complication of radiotherapy for head and neck tumours such as nasopharyngeal carcinoma (NPC)
- May be hypothalamic in origin - ↑ prolactin and diabetes insipidus often present
- ACTH deficiency- often asymptomatic and unmasked by stress

Hypogonadism

Hypopituitarism: Low oestrogen/testosterone; **FSH/LH normal/↓**

Gonadal failure: Low oestrogen/testosterone; **FSH/LH↑**
eg Turner's Syndrome; Klinefelter's Syndrome

Onset before puberty:

Female: Primary amenorrhoea, impaired breast development

Male: Small testes, penis and scrotum; ↓ hair (facial, pubic, axillary, body);
high-pitched voice, ↓ muscle mass & strength

Eunuchoidism if no growth impairment

(span>height, lower>upper segment); delayed epiphyseal closure

Onset after puberty:

Female: Secondary amenorrhoea

Male: ↓ libido; impotence; ↓ morning erection; infertility

↓ hair as above (eg ↓ shaving); failure of scalp hair to recede

Testes soft, size normal/mild↓; normal external genitalia

Tests of Anterior Pituitary Function

GH & ACTH deficiency - Gold standard:

Insulin Tolerance Test: ↓ GH and cortisol responses to hypoglycaemia
Contraindicated if epilepsy or coronary artery disease.

Alternative tests for adult GH deficiency: Glucagon test; Macimorelin test (oral non-peptide agonist of GH secretagogue receptor [Ghrelin receptor])

ACTH deficiency – ↓ cortisol response to 1 µg Short Synacthen Test **plus** normal or ↓ basal ACTH

(Impaired adrenal response to acute synacthen stimulation because of prolonged deprivation of tonic adrenal stimulation by ACTH)

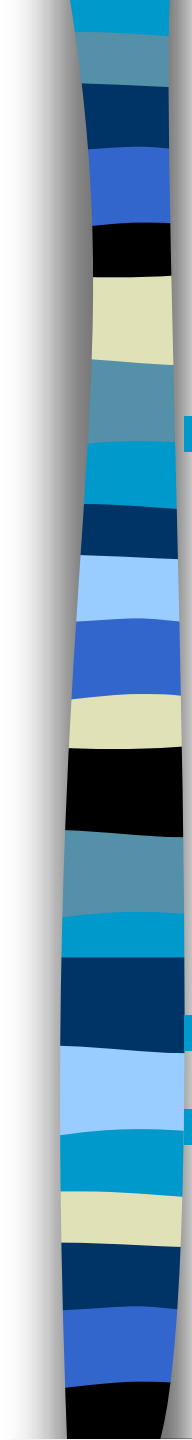
TSH deficiency- ↓ serum T4 with normal or ↓ TSH

TRH test: rarely indicated in daily clinical practice

LH/FSH deficiency- ↓ serum sex hormones; normal or ↓ FSH/LH

LHRH test: if diagnosis uncertain or fertility induction being considered

Hyperprolactinaemia - basal, non-stressed



Treatment of Hypopituitarism - Replacement Therapy

GH deficiency

- GH: children $\pm$ adults; daily subcutaneous injection
- Weekly sustained release GH approved by FDA

Gonadotrophin deficiency

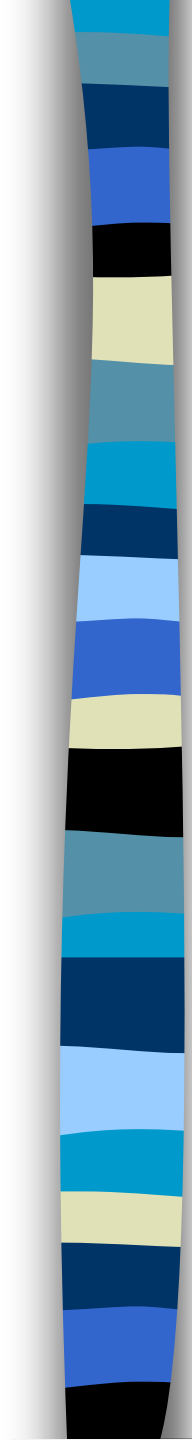
- Testosterone, oestrogen $\pm$ progestagen
- Gonadotrophins for fertility induction:
HMG/HCG (human menopausal gonadotrophin/human chorionic gonadotrophin); recombinant FSH/LH

TSH deficiency - thyroxine

ACTH deficiency - hydrocortisone (cortisol)

Diabetes Insipidus (DI): AVP-D or AVP-R

- Deficiency of, or resistance to, arginine vasopressin (AVP, also known as anti-diuretic hormone), leading to impaired ability to produce concentrated urine in response to $\uparrow$ serum osmolality or volume depletion
 - Presentation - polyuria, polydipsia $\pm$ hypernatraemia
 - Copious quantity of dilute urine
 - Dx: Inadequate rise in urine osmolality despite high serum osmolality basally or during water deprivation test
 - 2022: consensus proposal to replace the name Diabetes Insipidus with
 1. **Arginine Vasopressin Deficiency or AVP-D** (Cranial DI): Lesions of hypothalamus, pituitary stalk or posterior pituitary; urine osmolality $\uparrow$ with DDAVP (1-Deamino-8-D-arginine vasopressin) treatment
 2. **Arginine Vasopressin Resistance or AVP-R** (Nephrogenic DI): no response to arginine vasopressin or DDAVP; Genetic/acquired
- Reason for change in name: to eliminate confusion with diabetes mellitus and endorse a name that more accurately reflects the disease's etiology.**



F/40 Oligomenorrhoea for one year

- Interval of menstruation changed from 28 days to 70-90 days
- Serum prolactin (PRL) 1923 mU/L (N<500); low oestrogen; LH/FSH not ↑
- Normal fT4, TSH, 1μg Short Synacthen Test
- Visual field normal; no acromegalic/Cushingoid features
- MRI pituitary: tumour 2.1 cm (height); left suprasellar extension encroaching on optic chiasm
- Bromocriptine started: PRL ↓ 145 mU/L; menstruation returned
- MRI pituitary 4 months after bromocriptine: no tumour shrinkage
- Diagnosis: hyperprolactinaemia due to stalk compression by non-functioning pituitary macroadenoma
- Transsphenoidal surgery: tumour stained negative for all pituitary hormones
- Post-op diabetes insipidus and insulin tolerance test showed subnormal peak cortisol and GH; oral DDAVP and hydrocortisone 10 mg om required

TELL US WHAT

JC Whole Class Session

YOU THINK

THANK YOU!



Look for today's teacher
and the others you may have missed.
(HKU Portal login required)